Audio Plugin - User Manual & Technical Documentation

LLAMAJA REV5

Eng. David Rábago Félix
Audio Plugin Developer & Mechatronic Engineer

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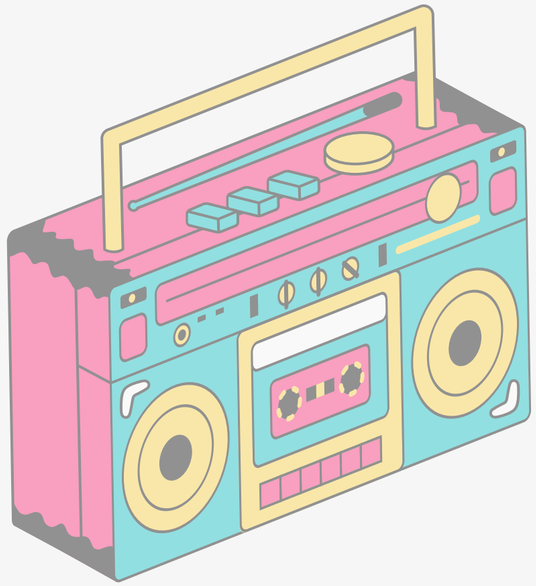

INTRODUCTION

Welcome to Llamaja REV5, a digital reverberation unit inspired by classic 80s hardware! This plugin combines vintage algorithms with modern FDN (Feedback Delay Network) architecture, providing incredibly dense, natural, and highly customizable spaces for your mixes.

Why Llamaja REV5?

- **FDN Architecture: 8x8 Hadamard matrix for exponential echo density.**
- **Vintage Inspiration: Early reflections based on classic Moorer/Schroeder designs.**
- **Total Control: Independent EQ for pre-reverb and wet signals.**
- **Creative Flexibility: 24 Factory Presets ranging from realistic halls to endless ambient tunnels.**

FEATURES OVERVIEW

1

🏛️ Advanced FDN Reverb Engine

- 8x8 Feedback Delay Network with lossless Hadamard mixing.
- Exponentially growing echo density for natural decay.
- Independent LPF (Low Pass Filter) per delay line mimicking real-room high-frequency absorption.

2

🔊 Independent Early Reflections (Moorer)

- 8 distinct reflection taps per channel.
- Prime-number-based delay times to prevent phase cancellation (comb filtering).
- Equal-power ER/Rev Balance control for perfect depth staging.

3

🎛️ Dual 3-Band EQ System and Visual Animation

- Pre-EQ to clean up the signal before it hits the reverb tank.
- Prog-EQ to shape the tone of the wet tail exclusively.
- Smooth Low-shelf, Mid-peak, and High-shelf bands.
- Spectrum pixel animation.

4

🔄 Pre-Diffuser & Stereo Modulation

- 4 Allpass filters in series to smooth transients.
- Independent LFO modulation (Sine/Cosine) for wide stereo decorrelation.
- Eliminates metallic ringing on percussive sources.

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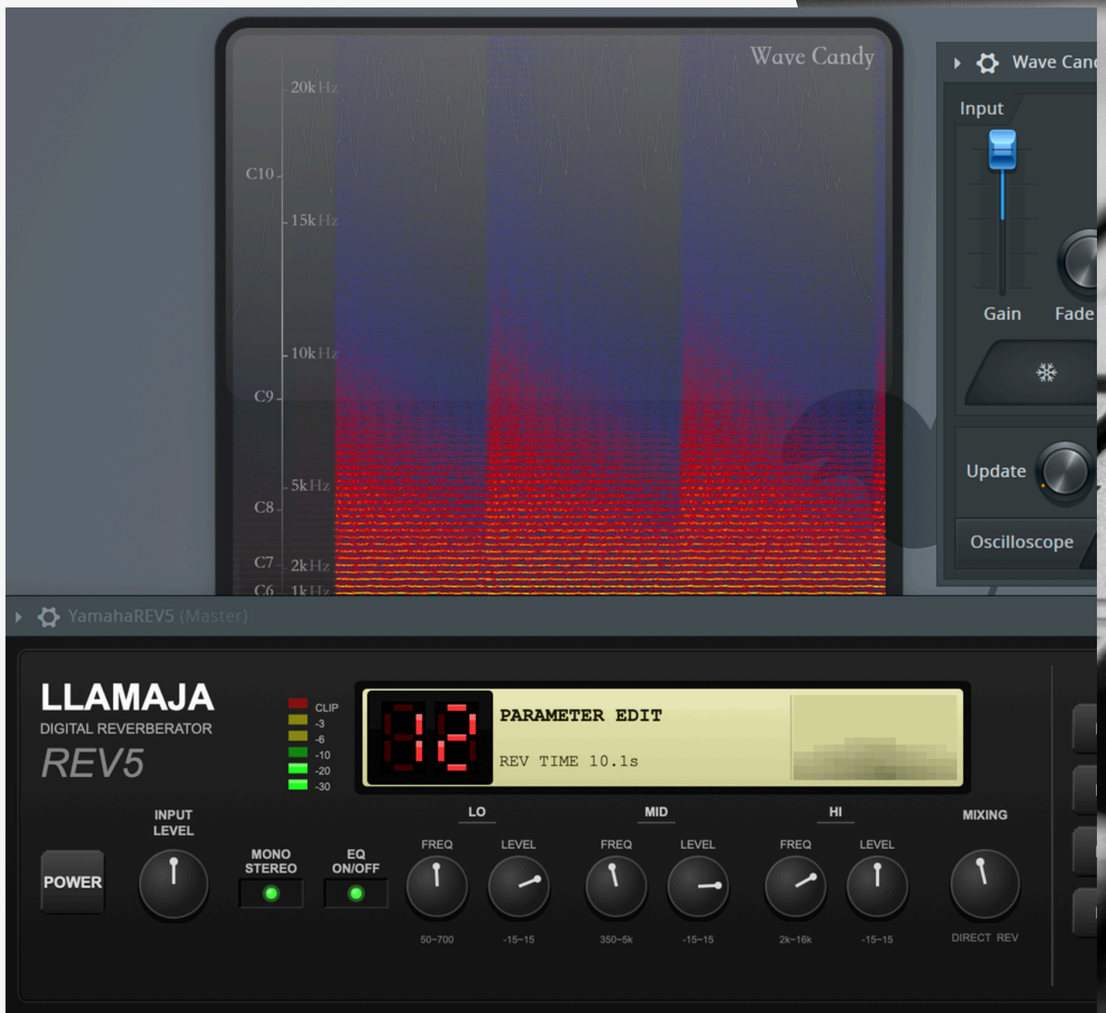

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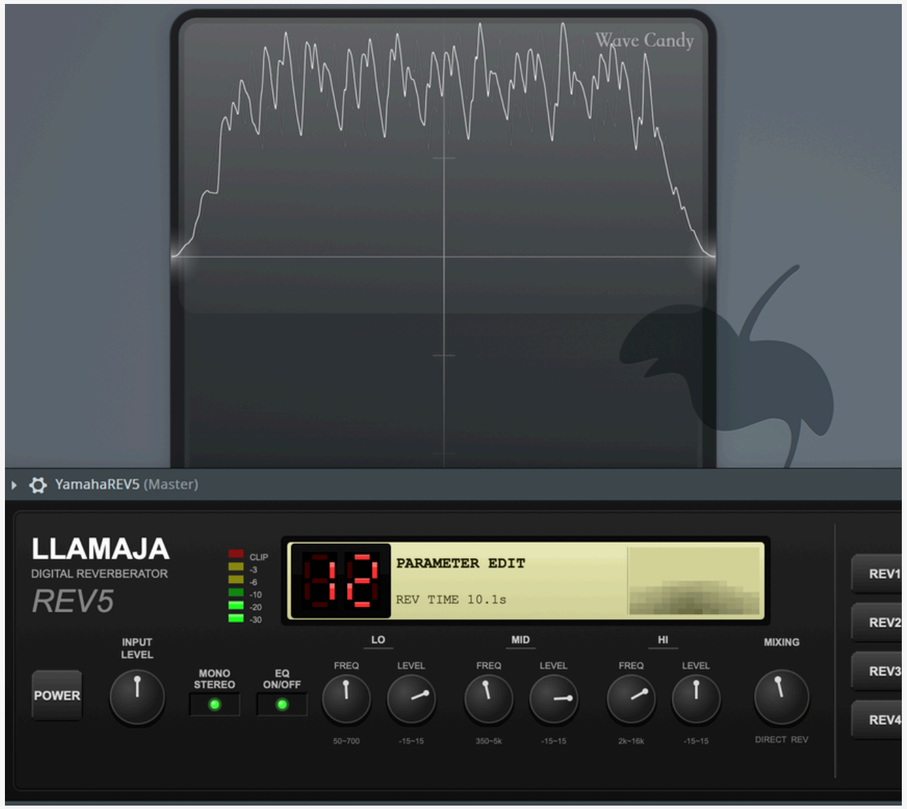

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INSTALLATION GUIDE

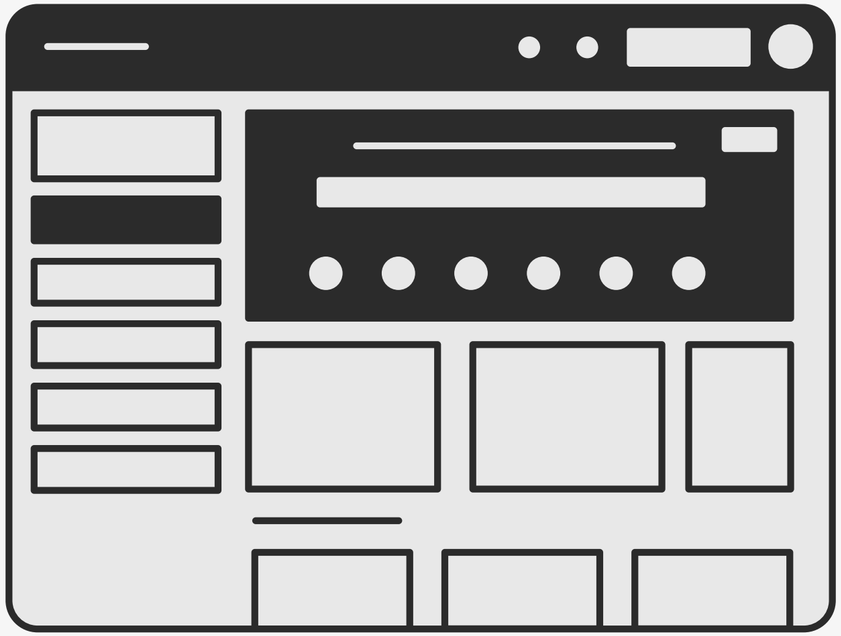
STEP 1: DOWNLOAD

Download the latest .zip package (Mac or Win64) from the official release page or repository.

STEP 2: INSTALL PLUGIN FILES

Move the plugin files to the appropriate directories:

- macOS: /Library/Audio/Plug-Ins/VST3 or /Components
- Windows: C:\Program Files\Common Files\VST3



STEP 3: RESCAN PLUGINS

- Open your DAW
- Rescan plugin directory
- Locate "LlamajaREV5" in your plugin list

STEP 4: ACTIVATE

Load the plugin on any audio track or bus

System Requirements

- Operating System: macOS 11.0 or later
- Architecture: Apple Silicon & Intel
- DAW Compatibility: All major hosts (Logic Pro, Fl Studio, Pro Tools, etc.)
- Plugin Formats: AU, VST3

INTERFACE CONTROL

Main Controls

📣 INPUT LEVEL
- Pre-reverb gain staging
- Range: -24dB to +24dB
- Prevents internal clipping

🎧 MONO/STEREO
- Processing mode selector
- Mono: compact/centered reverb
- Stereo: wide immersive field

🎛️ LO FREQ (50-700HZ)
- Low shelf frequency point
- Controls bass body in reverb
- Cleans mud when lowered

📈 LO LEVEL (-15 TO +15DB)
- Controls gain body in reverb
- Cleans mud when lowered

🎛️ MID FREQ (350HZ-5KHZ)
- Midrange parametric center
- Shapes vocal/instrument presence
- Critical character control

📈 MID LEVEL (-15 TO +15DB)
- Midrange gain adjustment
- Balances reverb body
- Prevents frequency masking

🎛️ HI FREQ (2K-16KHZ)
- High shelf frequency point
- Controls air and brightness
- Vintage vs modern tone

📈 HI LEVEL (-15 TO +15DB)
- High shelf gain amount
- Tail sparkle vs warmth
- Perfect for vocal plates

🌀 MIX (0-100%)
- Dry/Wet blend control
- 100% = aux bus usage
- 20-40% = insert blending

📊 DIRECT/REV DISPLAY
- Shows current dry/wet balance
- Visual mix feedback
- Confirms processing amount

INTERFACE CONTROL

Button Panel/Navigation

🔴 REV1, REV2, REV3, REV4

- Selects main reverb algorithm banks
- Quick access to different sonic characters
- Hardware-style preset switching

🔢 NUMERIC KEYPAD

- Direct value entry
- Preset/memory number selection
- Precise parameter adjustment

💾 MEMORY

- Opens preset management
- Shows current memory slot
- Factory/User preset browser

📚 OTHERS

- Secondary functions access
- Utility pages and advanced settings
- Less-used but essential controls

🟠 ER1, ER2

- Accesses Early Reflection variants
- Controls initial room reflections only
- Perfect for precise room sizing

💾 STORE

- Saves current settings to memory
- “Freeze” your reverb design
- Overwrites selected slot

👤 USER

- User preset bank access
- Your custom reverbs (8 slots)
- Separate from Factory presets

⬇️⬆️ DOWN/UP

- Navigate presets/parameters
- Incremental value adjustment
- Keyboard-style navigation

🔇 MUTE

- Silences reverb output only
- Dry signal still passes
- Perfect for mix verification

🟡 PARAMETER

- Selects which parameter to edit
- Prepares numeric keypad input
- Rack-style parameter navigation

📥 RECALL

- Loads saved preset/memory
- Instant A/B preset comparison
- No parameter reset needed

🎚️ EQ

- Opens EQ section (Pre/Program)
- Dual 3-band parametric control
- Pre-reverb and wet-tail shaping

🛠️ UTILITY

- System settings and diagnostics
- Global configuration
- Advanced user options

🟢 LEVEL

- Low shelf frequency point
- Controls bass body in reverb
- Cleans mud when lowered

🔧 INITIAL

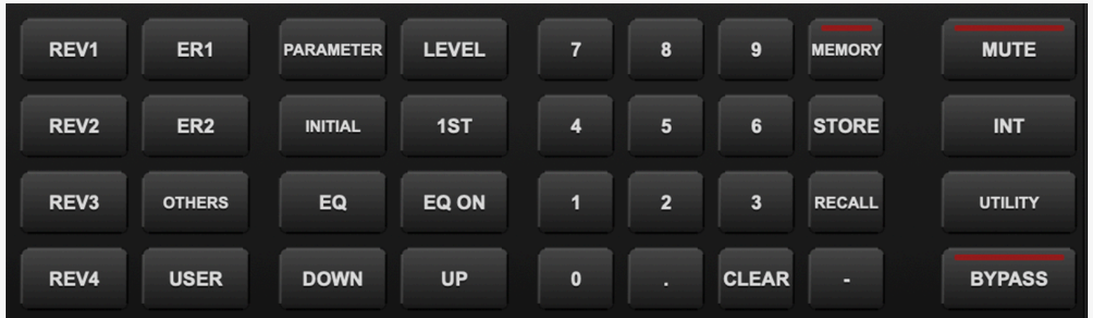
- Returns to factory default state
- Clean starting point for design
- Resets all parameters

✅ EQ ON

- Toggles all EQ sections
- Compare EQ'd vs raw reverb
- Instant tonal verification

⛔ BYPASS

- Complete plugin bypass
- 100% dry signal through
- Essential A/B comparison tool



PRACTICAL APPLICATIONS

VOCAL PROCESSING 🎤

- Lush vocal plates and ballads (Presets F14, F15)
- Adding width without washing out the lead
- Creating depth using longer Pre-Delays (30ms+)

DRUM BUS ENHANCEMENT 🥁

- Snare drum thickening (Preset F08 Room 2)
- Overhead cohesion using short room algorithms
- 80s style Gated Reverb effects (Preset F20)

GUITARS & SYNTHS 🎸

- Endless ambient soundscapes (Presets F19 Tunnel, F20 Canyon)
- Chorus Hall for wide stereo lead tones (Preset F24)
- Adding natural acoustic space to DI instruments

TECHNICAL SPECIFICATIONS

AUDIO PROCESSING

- **Sample Rates:** 44.1kHz to 192kHz
- **Bit Depth:** 32-bit floating point internal processing
- **Channels:** Stereo In / Stereo Out (True Stereo processing)
- **DC Blocker:** Built-in 1-pole high-pass to prevent DC offset accumulation on long tails.

TECHNICAL FEEL

- **Sound density:** High, with fast build-up of reflections.
- **Spatial realism:** Strong due to separate ER and late-tail processing.
- **Control range:** Wide enough for subtle ambience or extreme creative reverb.
- **CPU profile:** Intended to remain efficient for everyday mixing.

PLUGIN SPECIFICATIONS

- **Framework**: JUCE 7/8
- **Code**: C++17
- **DSP Architecture**: 8x8 Hadamard FDN, Moorer ER, Schroeder Allpass
- **Parameters**: AudioProcessorValueTreeState managed with smoothed values to prevent zippering.

TECHNICAL SPECIFICATIONS

FX SPECIFICATIONS

- **Type:** Algorithmic digital reverb.
- **Core topology:** 8×8 Feedback Delay Network (FDN) with Hadamard mixing.
- **Early reflections:** Independent Moorer-style ER stage.
- **Diffusion:** Multi-stage allpass diffusion network.
- **Tone shaping:** Dual 3-band EQ system, pre-reverb and wet-tail EQ.
- **Modulation:** Stereo LFO modulation for diffusion/decorrrelation.
- **Dynamics:** Smoothed parameter handling to avoid zipper noise.
- **Low-end control:** Internal filtering to reduce muddy buildup in long tails.
- **Stereo behavior:** Stereo widening and decorrelation in the late reverb field.
- **Bypass path:** True bypass-style plugin bypass at the host level.
- **Latency:** Designed for real-time use with no intentional additional latency.

AUDIO CHARACTERISTICS

- **Tail character:** Dense, smooth, and highly controllable.
- **Space behavior:** From compact rooms to large halls and ambient washes.
- **Coloration:** Can go from clean and modern to darker and vintage-style.
- **Presence control:** Early reflections and tail balance are independently adjustable.
- **Mixing style:** Works well both on aux sends and direct insert chains.

TROUBLESHOOTING

Reverb Tail Sounds Muddy/Boomy

- Activate the PRE-EQ and apply a Low-Shelf cut at 200Hz.
- Decrease the LF Ratio.

Audio Clipping / Distortion

- Lower the Input Level knob (-3dB to -6dB).
- Check your DAW's channel gain staging.

Plugin Not Appearing

- Confirm correct installation path
- Rescan plugins in your DAW
- Check plugin format compatibility

CPU Usage High

- LLAMAJA REV5 is optimized for low CPU usage
- If issues occur, check other plugins in session
- Increase your DAW's audio buffer size (e.g., 256 or 512 samples).

Support Contact

For technical support or bug reports:

- **GitHub Issues**: github.com/DavidSignals/LlamajaRev5/issues
- **Email**: david.rabago.felix@gmail.com
- **Documentation**: Full documentation available on GitHub

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David Rábago Félix

Professional Alias: David Signals

Background

Audio plugin developer and mechatronic engineer with a passion for creating tools that enhance creative workflows. Specializing in JUCE framework development and digital signal processing. My hands-on hardware expertise includes designing guitar pedal effects and building synthesizer circuits with integrated circuits and programmable platforms like ESP32, Raspberry Pi, Arduino, Teensy 4.1, and Daisy Seed. Currently designing a multi-effects pedalboard. Guitarist and music lover.

Philosophy

"Great tools should be invisible, they should empower creativity without getting in the way. Mono/Stereo Ears embodies this principle through its intuitive design and reliable performance."

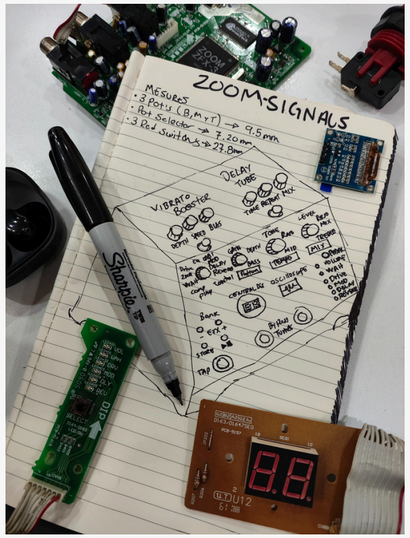
Connect

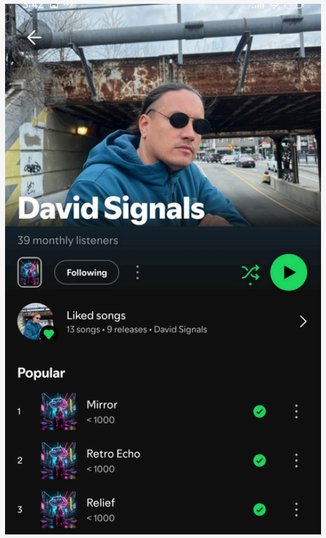
- GitHub: https://github.com/DavidSignals
- LinkedIn: www.linkedin.com/in/davidrabagofelix

Acknowledgments

Special thanks to the JUCE framework team and the audio developer community for their continuous inspiration and support.

ABOUT ME





Eng. David Rábago Félix
“David Signals”

📞 *SUPPORT & RESOURCES*

CELLPHONE

+52 6673091785

EMAIL

david.rabago.felix@gmail.com

SOCIAL NETWORKS

@davidsignals
@_deivip_
https://github.com/DavidSignals